

tem's graphical user interface. Imagine a speech user interface that gives you one single method of activating the device and then all its functions are accessible through the speech command of the user and the voice feedback of the device. There should be no need to use a screen, mouse or keyboard! Further, a real speech user interface should allow third-party applications to extend their functions to other programs through a series of application programming interfaces.

In a recent review, mobile expert Adam Z. Lein commented that the latest Xbox 360 offered something close to this—if not a complete platform, at least an extensible speech interface. The device appears to have a consistent speech interface that is shared by all applications and games that have been programmed to make use of it.

"No matter if I'm in the Dashboard, the YouTube app, Netflix, Crackle, MSNBC or Star Wars Kinect, if I say 'Xbox' it will start listening to me and highlight the relevant commands that I

can say. I can tell the TV what I want it to do instead of picking up a plastic controller and pressing buttons. Plus, since the speech user interface is shared throughout the system, apps like YouTube and Netflix don't have to reinvent the wheel in order to get proper voice control support," writes Lein.

"Unfortunately, the speech interface of Xbox still requires your eyes on the screen in many situations since it does not provide voice feedback for any command. Hopefully, we will finally see some real forward movement in



LG magic remote control

## Carmakers also focus on user interface

The new models of Cadillac focus on the so-called Cadillac user experience or CUE. Now Cadillac users can put away their phones or tablets, yet channel the devices' capability and media into the car using an elegant and intuitive user experience.

- The CUE innovations are primarily focused on proximity sensing, natural voice recognition, haptic feedback and a unique capacitive touchscreen.
- Proximity sensing minimises the display when not in use.
- The haptic feedback gives users a pulsing sensation when they select menu items.
- The 8-inch capacitive touchscreen allows for swipe and pinching gestures to move and resize items on the main screen.
- The built-in 3D navigation includes turn-by-turn natural speech recognition and auto-fill location input.

The CUE designer team has also opened up the Linux-based CUE platform to developers, to ensure that the CUE stays ahead of tech trends always.

the area of speech user interfaces for Windows Phone 8 and Windows 8 since that seems like the next step for human-computer interactions of the future—considering we already have 3D gesture recognition with Kinect," adds Lein.

## Why not simply think?

While voice and gesture control are still in a nascent stage, products for thought-based user interfaces are already hitting the market. These too are still very, very nascent but nevertheless there is an option for applications or people with special needs.

One interesting product in this spectrum is inventor Tan Le's EPOC headset, developed and marketed by Emotiv Systems. The Emotiv EPOC is a wireless neuro-headset that taps the latest developments in neuroscience to enable high-resolution neuro-signal acquisition and processing. It is a decent-sized wearable headset that uses electroencephalogram (EEG) readings to collect and decipher the user's thoughts and translate them into instructions.

Users can control and influence the virtual environment with their mind, access applications and play games developed specifically for the EPOC. Or, they can use the EmoKey to connect to current PC games and experience them in a completely new way. The EEG readings can be used by researchers for neuro-therapy, biofeedback, and brain computer interface study and design. A software development kit is also available for developers to

come out with interesting applications using EPOC.

The Sekati brain-computer application programming interface is another interesting tool. The framework allows you to interface with an EEG device via a socket (provided by the open ThinkGearConnector server; part of NeuroSky's open development SDK) and allows you to read connection status, signal strength, NeuroSky's proprietary algorithmic eSense values as well as raw EEG values. The tool may be used for medical observation, games or applications.

While NeuroSky and Emotiv are the two main commercial competitors, there are other players such as Cyberkinetics, Mindball and Starlab too which make brain-computer interface products.

In short, there is a lot of interesting work happening in the space of user interfaces—whether based on speech, touch, gesture or thought. However, designers have to keep in mind that it is not just the technology but the usability and user-engagement that are important for any user interface.

"A user interface must address what first-century BC architect Vitruvius stated—architecture (read 'all creative design') must address commodity, firmness and delight: value, structure and aesthetics," reminds Ranjit Makkuni, president-Sacred World Foundation ([www.sacredworld.com](http://www.sacredworld.com)), multimedia researcher, designer and musician. ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai